Section B

Answer **one** question from this Section in the spaces provided.

6 **(a)** The energy levels of a hypothetical one-electron atom are given by

$$E_n = -\frac{27.90}{n^2} \text{ eV}$$

where $n = 1, 2, 3, \ldots$

(i) Calculate the energies of the four lowest energy levels and construct a clearly labelled energy level diagram.

[3]

(ii) If the atoms are in the ground state and are bombarded by electrons of kinetic energy 26.5 eV, determine the highest energy level that an atom can reach. Show your working clearly.

highest energy level = ... [3]

(iii) Indicate, with arrows, in your energy level diagram in **(a)(i)** all the possible transitions that produce emission lines when these atoms de-excite.

[2]

(iv) Calculate the longest and the shortest wavelengths of the photons emitted during these transitions.

longest wavelength = ... nm

shortest wavelength = ... nm [2]

(b) In a modern X-ray tube, electrons are accelerated through a large potential difference and the X-rays are produced when electrons strike a metal target embedded in a large piece of copper.

Fig. 6.1 shows an energy level diagram of an atom of a hypothetical metal.

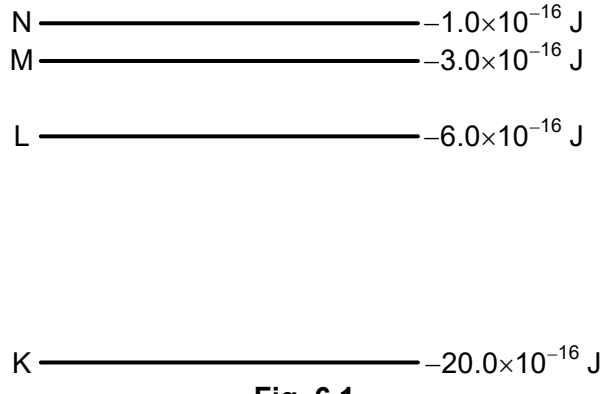

Fig. 6.1

The emission spectrum of the metal when it is bombarded by a beam of fast-moving electrons is shown in Fig. 6.2.

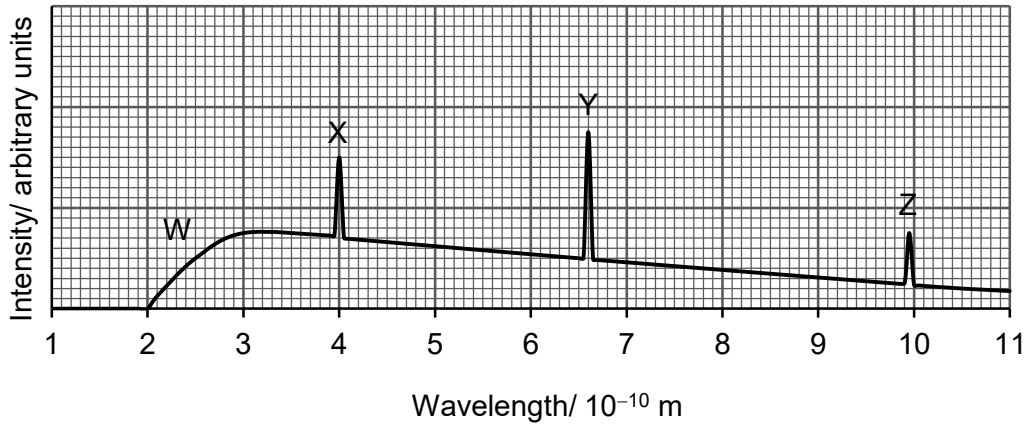

Fig. 6.2

(i) Describe the processes that occur at the target which produce the continuous spectrum W.

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..... [4]

(ii) Calculate the accelerating potential of the X-ray tube.

accelerating potential = ... V [3]

(iii) Using Fig. 6.1, deduce which of the lines X, Y and Z in Fig. 6.2 correspond to L_α, L_β and M_α lines.

L_α = ...

L_β = ...

M_α = ... [2]

(iv) Explain why the K spectral lines are missing from Fig. 6.2.

.....

..... [1]